EMMANUEL NKAKA

Computer Science Student | Kitwe, Copperbelt Province, Zambia

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PROFILE

Third-year BSc Computer Science student at the Copperbelt University with hands-on experience across software, systems and hardware: machine learning for IoT security, operating systems in C/C++, GUI programming, web development, and Arduino-based electronics and circuitry. Interested in applying technology to development challenges in Zambia. Seeking an internship or attachment where I can build, debug and learn.

EDUCATION

BSc Computer Science

Copperbelt University (CBU), Kitwe, Zambia Currently in Year 3

Coursework: Operating Systems, Numerical Analysis, Data Communications & Networking, Machine Learning, Software Engineering, Object-Oriented Programming (Java).

TECHNICAL SKILLS

Programming: C++, Java, Python, JavaScript, PHP, HTML, CSS, SQL (MySQL)

Electronics & Embedded: Arduino programming, circuit design and breadboarding, sensor and component interfacing, circuit troubleshooting

Machine Learning: Data preprocessing, model training and evaluation, intrusion/anomaly detection (UNSW-NB15 dataset)

Systems & Tools: Linux (Kali), Git/GitHub, MSYS2 MinGW, Code::Blocks, Makefiles, GTK3

Web: Responsive HTML/CSS/JS portfolio sites; PHP/MySQL backends

Computer repairs: experience with Mac-os and Windows system

PROJECTS

IoT Intrusion Detection System (group project) 2026

- Built a machine learning pipeline to detect malicious network traffic in IoT environments using the UNSW-NB15 dataset.
- Worked in a team of four covering data preparation, model selection, evaluation and documentation.

Arduino & Circuitry Projects 2025

- Designed, wired and programmed Arduino-based circuits, interfacing sensors and components and debugging hardware and code faults.

MiniOS: Scheduler & Process Management Coursework

- Implemented CPU scheduling algorithms, inter-process communication (IPC), memory management and logging in C/C++.

News Webpage Web development

- Developed a news website with a PHP/MySQL backend, covering content management and page design.

GTK3 GUI Application C programming

- Built native GTK3 windowed applications in C, configuring the Code::Blocks and MSYS2 MinGW toolchain.

HTML/CSS Portfolio Website Web development

- Designed a modular portfolio with scroll navigation, glow-effect buttons and varied page layouts (blog, gallery, quiz, e-commerce).

STRENGTHS

- Systematic troubleshooting: Makefile and compiler errors, header issues, toolchain configuration, hardware and circuit faults.
- Comfortable moving between software and hardware, from high-level ML to, circuits.

INTERESTS

IoT and machine learning, operating systems, embedded electronics, GUI programming, soccer, anime and music.